Taiwo Alabi

Data Scientist. I build AI systems end-to-end and publish exactly where each one fails

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Profile

I tune intelligence to the cost of being wrong. Before I train anything I establish what the problem looks like solved badly: the majority-class score, the simple rule, the off-the-shelf tool. A model that cannot beat those does not deserve to exist, and one that does has earned the number it reports. I learned why that matters on a telecom platform carrying 400 million usage records a day, where a bad deploy meant fourteen downstream systems read nothing, or read twice. I am a Master's student in Artificial Intelligence with five systems designed and produced end to end, one deployed and publicly reachable, and I can explain every one of them to you during our interview.

Professional Experience

Turing, Remote (Dublin) Apr 2026 to May 2026
AI Quality Analyst, Google Gemini evaluation programme

- I ***measured*** the quality of ***Google Gemini*** responses on ***personalization*** features, then wrote prompts to test the models, and documented the process for a ***Reinforcement Learning from Human Feedback (RLHF)*** project.

Huawei Technologies (via RGS), Globacom Telecommunications department, Lagos Summer 2024 & 2025
Product Manager Intern

- Problem: On a national mobile network in one of ***Africa's largest mobile markets, 400 million daily call records represent 400 million bills*** that real people pay.
- Action: I designed and tested the API layer for a cloud integration between the contact centre platform and a commercial bank's customer care line. I tracked every interface the bank had to expose against what we had actually received, and tested each one with curl before it went near a voice prompt.
- Result: When authentication fails, the call does not retry silently and does not dead-end. **It transfers to a person.** That is the rule I have built into every system since.
- I also worked with the solutions architect on a data migration that partitions customer data into domain-specific buckets, and with the test team designed over 440 User Acceptance Test (UAT) cases. During the phased cutover a downstream storage target filled and blocked uploads, and a control connection reset stopped one CDR stream. Both surfaced in verification, before any customer saw them.

Selected Projects

Smart Medical Records Engine FastAPI, PostgreSQL/pgvector, LangGraph, Docker

- **Problem:** A doctor has ***ten minutes*** and a patient history spread across ***years of notes***, the one critical line, ***a past reaction to a medicine,*** gets missed.
- Action: A general vector search across a hospital corpus will eventually surface the wrong patient's record, so scoping happens at the query layer rather than as a filter applied afterwards. Thirty adversarial probes, zero leaks. Self-hosting Llama 3.1 costs accuracy against a frontier model, but patient records leaving the network is not a trade a hospital can make.
- **Result:** I ran the ***whole stack***, the API, ***PostgreSQL with 768-dimensional vector search*** and a ***self-hosted Llama 3.1***, across ***four Docker Compose services***, so ***patient data never leaves the hospital's own network***.
  Read the write-up (PDF) | Architecture diagram | Data model | Source code

Customer Care Process Automation Python, scikit-learn, BPMN 2.0

- **Problem:** Hundreds of customer complaints used to get ***sorted by hand every morning***, which means we used to have more misrouted tickets because of ***human errors***. Customers ***waited for minutes*** until proper routing directed their ticket to the right back-end team, followed by a few minutes of communication with that customer to resolve the issue.
- Action: I modeled the process in Business Process Model and Notation first, then split it: a rule-based bot enforcing field-level conformance against a standard report catalogue, the kind of signed field definitions I

worked to at Huawei, and an AI agent reading customer complaint tickets and routing them to the appropriate back-end team. It reached 95.2% routing accuracy on held-out tickets, 40 of 42, a corpus small enough that I report the count and not just the rate.

- Result: The decision was the gate, not the classifier. Below 0.55 confidence the agent escalates to a human rather than forcing a label, and I report the lower automation rate that produces. Naive Bayes was chosen over heavier models because 210 tickets will overfit anything larger, and because a misroute has to be inspectable.

  Read the write-up (PDF) | BPMN process models | Confusion matrix | Source code

TermsGuard AI Sentence transformers, FAISS, Claude API, Flask

- Problem: A 2008 study estimated that reading every privacy policy an average internet user meets would take about 244 hours a year. A 2017 Deloitte survey of 2,000 consumers found 91% accept terms without reading them.
- Action: The failure mode was hallucination, so every flag is grounded in GDPR text retrieved from a 4,527-chunk index rather than in the model's impression of it. Cosine similarity alone marks "we do not sell your data" as a violation, because negation barely moves an embedding, so explicit rule guards sit on top of the similarity score rather than instead of it. Invented content fell 77% and F1 reached 87% on a 205-clause reference set.
- **Results:** I cut ***end-to-end latency from 28 seconds to under 10***, and added a ***Claude verification stage*** that removes false positives, and built a cosine-only fallback when no API key is set.

  Live demo | Read the write-up (PDF) | Source code

Automated QA Gate for AI Generated Imagery TensorFlow/Keras, ResNet50, scikit-learn

- **Problem:** ***AI-generated fashion campaign images*** silently drift over multiple prompt generations; the character's ***identity starts to shift*** or specific details I want to retain become inconsistent**.**
- Action: Before building anything I ran two baselines. Pixel difference reached F1 0.59, k-nearest neighbours 0.26. The Siamese network on a frozen ResNet50 reached 0.90 with precision 1.00 and zero false positives on a 90-pair test set. That gap is the result; 0.90 on its own is not. I moved the decision threshold from the sigmoid default of 0.5 to 0.4821 on validation data only, because a rejected good image costs one regeneration and an accepted bad image reaches the client.

  Read the write-up (PDF) | ROC curve and confusion matrix | Architecture diagram | Source code

Education

National College of Ireland, Dublin Jan 2026 to Jan 2027
MSc Artificial Intelligence (part time)

Modules: Machine Learning, ***Intelligent Agents and Process Automation, AI-Driven Decision Making, Engineering and Evaluating AI, Natural Language Processing, Data Governance and Ethics,*** Database Programming, Foundations of AI.

Afe Babalola University, Lagos Sep 2021 to Feb 2025
BSc Computer Science, Second Class Honours, Upper Division (2.1)

Technical Skills

Languages: Python, SQL

Data & Infrastructure: Hadoop HDFS, HBase, GaussDB, PostgreSQL, pgvector, Docker, Docker Compose, Kubernetes, FastAPI, Flask, Google Cloud

Machine Learning: scikit-learn, pandas, TensorFlow/Keras, transfer learning, reinforcement learning, Q-Learning, neural networks, Transformers

AI & Agents: RAG pipelines, vector search (pgvector, FAISS), LangGraph, LLM evaluation, genetic algorithms

Data Governance: GDPR data cataloguing, data lineage, data quality validation, UAT design

Tools: Git, GitHub, Claude Code, Linux